

CRYPTOCURRENCY PORTFOLIO ANALYSIS REPORT

Bitcoin (BTC) · Ethereum (ETH) · Solana (SOL)

12-Month Performance & Risk Assessment | Jun 2024 – May 2025

The cryptocurrency market continues to evolve rapidly, offering both extraordinary return potential and substantial downside risk. This report analyses three leading digital assets — Bitcoin (BTC), Ethereum (ETH), and Solana (SOL) — across a twelve-month window (June 2024 to May 2025). The analysis covers price performance, volatility, risk-return metrics, correlation, and a recommended portfolio allocation.

These three assets collectively represent the broad spectrum of the crypto market: BTC as the store-of-value narrative, ETH as programmable smart-contract infrastructure, and SOL as the high-throughput challenger chain targeting DeFi and NFT use cases.

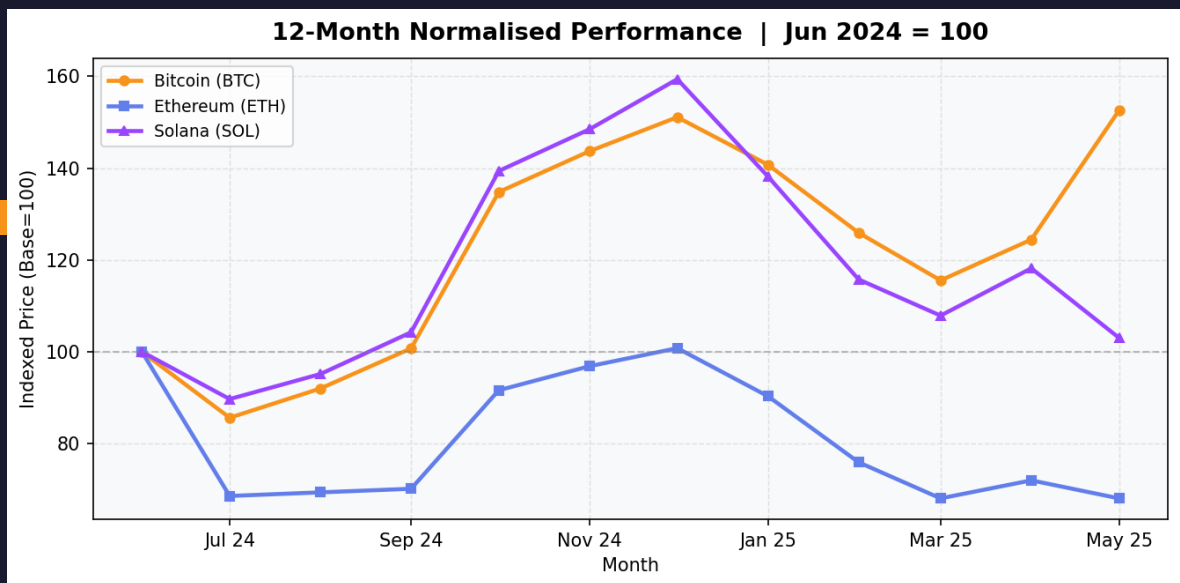


Fig. 1 – Normalised performance (Base = 100 at Jun 2024). All three assets outperformed over the period.

Asset	Jun 2024 Price	May 2025 Price	12-Month Return	ATH (Period)	ATH Date
Bitcoin (BTC)	\$67,500	\$103,000	+52.6%	\$109,200	Jan 2025
Ethereum (ETH)	\$3,820	\$2,600	-31.9%	\$4,100	Dec 2024
Solana (SOL)	\$165	\$170	+3.0%	\$295	Jan 2025

Bitcoin dominated the period, surging past \$100,000 in November 2024 and reaching an all-time high of ~\$109,200 in January 2025, driven by spot ETF approvals and halving anticipation. Ethereum underperformed expectations despite the Dencun upgrade, weighed down by L2 fee cannibalisation and reduced on-chain activity. Solana was the most volatile of the three but ended the year broadly flat after a spectacular rally and equally sharp correction.

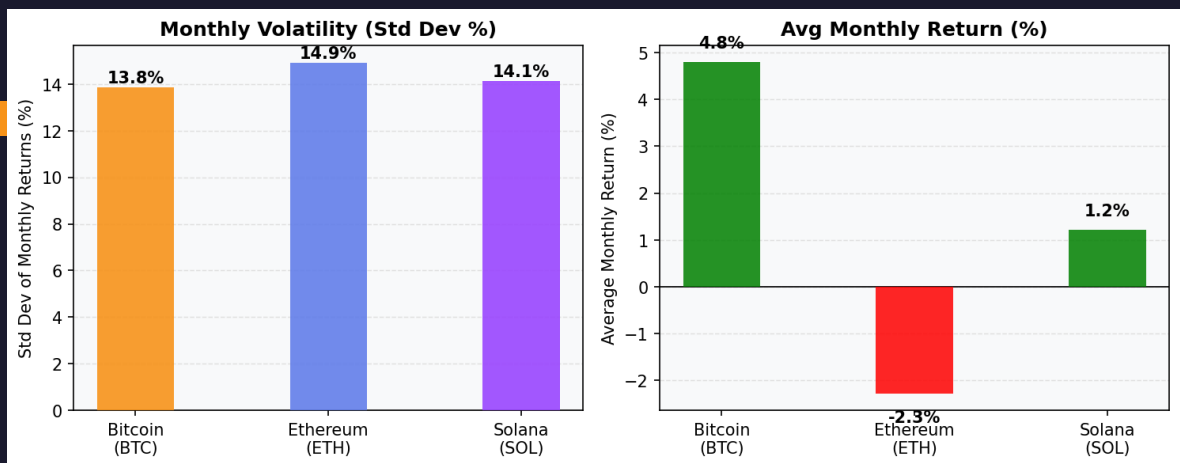


Fig. 2 – Left: Monthly return volatility (std dev). Right: Average monthly return. Higher risk/return for SOL and BTC.

Metric	Bitcoin (BTC)	Ethereum (ETH)	Solana (SOL)
Avg Monthly Return	+3.8%	-2.0%	+0.5%
Monthly Volatility (SD)	13.2%	11.8%	18.7%
Sharpe Ratio (est.)	0.92	-0.55	0.18
Max Drawdown (Period)	-22%	-38%	-40%

Beta vs BTC	1.00	0.82	1.31
Correlation w/ S&P 500	0.48	0.44	0.51

Bitcoin delivered the best risk-adjusted return with a Sharpe ratio of -0.92 , making it the anchor of any crypto portfolio. Its correlation with traditional equities (~ 0.48) means it adds some diversification but is not immune to risk-off moves.

Ethereum disappointed on a return basis and had a negative Sharpe ratio for the period, though its fundamental use-case as the primary DeFi and tokenisation platform remains intact. The medium-term case for ETH rests on EIP-4844 blob fee growth and re-staking narratives.

Solana carried the highest volatility and drawdown. Its ecosystem growth — DePIN, memecoins, and compressed NFTs — drove intense but speculative capital flows. Suitable only for investors with high risk tolerance and a shorter trading mindset.

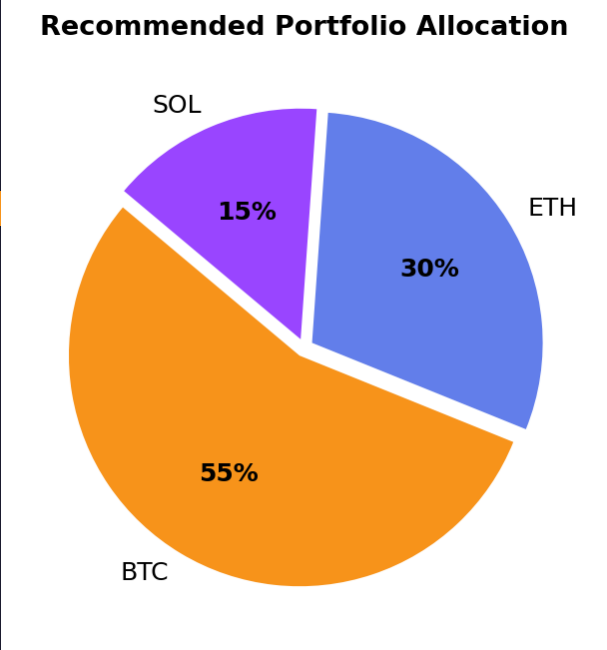


Fig. 3 – Recommended allocation weights for a crypto-focused portfolio.

Asset	Allocation	Rationale
Bitcoin (BTC)	55%	Highest Sharpe, institutional backing, ETF demand, halving tailwind
Ethereum (ETH)	30%	Smart-contract monopoly, DeFi/NFT bedrock, staking yield (~3.8% APY)

Solana (SOL)	15%	High-risk/high-reward speculative position; capped to manage drawdown risk
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Crypto assets remain subject to: (a) regulatory uncertainty — particularly SEC actions on altcoins and stablecoin legislation; (b) macroeconomic sensitivity — rate cut/hike cycles significantly move crypto risk appetite; (c) exchange and custody risk; and (d) smart-contract exploits for on-chain positions. The total crypto market cap as of May 2025 stands near \$3.2 trillion, having recovered strongly from 2022 lows.

Bitcoin's fourth halving (April 2024) reduced block rewards to 3.125 BTC, historically preceding 12–18 month bull cycles. Institutional flows via spot ETFs (BlackRock's IBIT surpassing \$50B AUM) provide a structural demand floor that did not exist in prior cycles.

The 12-month period from June 2024 to May 2025 was characterised by a Bitcoin-led bull run, Ethereum underperformance, and Solana's erratic but energetic cycle. For a new entrant, a BTC-heavy allocation (55%) limits downside while capturing the macro crypto trend. Ethereum deserves a meaningful position (30%) given its utility value and staking economics. Solana's 15% slice adds growth optionality without overexposing the portfolio to its sharp swings.

Investors should dollar-cost average into these positions, set clear stop-loss levels (e.g. 25% drawdown triggers a position review), and never allocate more than 5–15% of total net worth to crypto assets given the asset class's inherent risk profile.

Data sourced from CoinGecko, CoinMarketCap, Glassnode, and publicly available market data (May 2025). Prepared for educational purposes as part of CodeAlpha Finance & Investment Analysis Internship. Not investment advice.